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# Self-organized Speculation Game for the spontaneous emergence of financial stylized facts

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ABSTRACT

We sophisticate the Speculation Game to relax the need for rigorous tuning of a part of parameters for the emergence of characteristic statistics of price returns. Named as Self-organized Speculation Game (SOSG), thanks to the concept of the Bak–Tang–Wiesenfeld (BTW) sandpile model, the model can spontaneously adjust the number of market participants during the process of reaching the quasi-critical state. While the original Speculation Game's high reproducibility of financial stylized facts is maintained, the behavioral characteristics of this modified model, such as the dynamics of system size and the observations of power-law phenomena, strongly resemble those of the BTW model. The market size is evolutionally settled with fluctuations within a certain range, similar to the total sand grains behave in the sandpile model. SOSG infers the possibility that, in the real financial markets, it could be the self-organized quasi-criticality which works behind the spontaneous emergence of universal financial stylized facts.

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## 1. Introduction

### 1.1. Background

Time series of price returns in financial markets are known to have several quantitative and qualitative statistical properties collectively called stylized facts. Cont reported 11 representative statistical properties [1] including the well-known volatility clustering, which is characterized by the tendency for the large-scale price returns to cluster together [2–4]. Those features are quite nontrivial because they can be observed universally in different markets and instruments.

In the studies of econophysics, several emerging mechanisms for volatility clustering have been proposed based on bottom-up simulations employing simple agent-based models. The most popular, and maybe convincing one as well, is the effect of herding behavior [5,6], which refers to the intermittent synchronization of actions among traders. For example, based on studies with the family of Minority Game [7] typed models, one may infer that either a combination of inductive learning with dynamic capitals [8,9] or a performance-dependent switch between active and inactive strategies [3,10], could trigger the herding behaviors among the players and result in large-scaled intermittent price

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fluctuations. Another plausible mechanism is the order-book price formation. Typically, the Maslov model [11] suggests that the continuous double auction system alone can cause the clustering of volatile price fluctuations. As the latest study, we recently proposed alternative market scenarios with some empirical evidence in the study of the Speculation Game [12]. Specifically, it is found that heterogeneous holding periods in round-trip trades may agitate volatilities by broadly differentiated trading volumes caused by the diversification of capital among traders [13].

Despite these advancements in revealing the potential mechanism for volatility clustering, little is known about the *spontaneous* emergence of stylized facts in the financial markets. Stylized facts can be commonly observed irrespective of the types of markets, the lengths of histories, the differences in regulations, and so on. For instance, similar statistics of price returns can be observed either in the mature markets with histories of more than 100 years (such as US markets) or in the emerging markets with at most 30 years of history (such as Chinese or Indian markets). The inference of such a universal spontaneity is that the number of parameters of market models that need to be artificially tuned should be very small. In other words, a certain kind of spontaneous adjustment mechanism should work in the background. However, even the most recent market models, including the Speculation Game, essentially require the external fine-tuning of all control parameters for the reproduction of stylized facts.

In econophysics, one possible approach to addressing this problem is to conceptually assume the market system must stay in a nearly critical state [6,14–18], which might be supported by two primary aspects of stylized facts. First, the empirical (complementary) cumulative distributions of positive and negative price returns display power-laws in the tail part, with a finite exponents $\alpha$ ($2 < \alpha < 5$) [2,19,20]. Second, the autocorrelation of volatilities decays slowly with the time interval, which may be fitted either by a power-law [1,19] or by a logarithmic function [21]. In physics, criticality is defined without ambiguity as the state of a system in which the frequency distribution of some event demonstrates a power-law tail. Simultaneously, spatial and/or temporal correlation functions should also decay with a power law, indicating the existence of infinite correlation lengths [22]. On the other hand, the decay of autocorrelation of volatilities in some markets or instruments follows the logarithmic function rather than the power law [23]. Hence, it should be more appropriate to categorize the actual state of financial markets as the quasi-critical state instead of a pure critical one.

Regarding the process approaching the critical or quasi-critical states, there is a unique physical principle called self-organized criticality (SOC) that allows self-adjustment of a parameter corresponding to the system condition[1] [24,25]. In 1987, Bak, Tang, and Wiesenfeld (BTW) proposed the SOC concept elaborated with the famous sandpile model as a non-equilibrium system that can reach the critical state under quite general conditions [26]. In a two-dimensional BTW sandpile model, a sand grain falls on a randomly chosen site in an $L \times L$ lattice which causes a unit increase of the slope of that site. If the local value of slope got larger than a threshold, a landslide would occur causing the increase of slope in the neighbor sites. A chain of those landslides brings about an avalanche, through which sand grains roll down to the edge and fall off the lattice. Even such a set of simple rules can render the system critical, that is, the emergence of a power-law relationship between the frequency and size of avalanches. Note that there are SOC systems that can only reach quasi-critical states, such as the Olami–Feder–Christensen (OFC) earthquake model [27]. Consequently, the behavior of such a system is categorized as self-organized quasi-criticality (SOqC) [28].

To the best of the authors' knowledge, there are few agent-based financial market models equipped with the SOC setup. The Alfi–Pietronero–Zaccaria (APZ) market model [29–31], in which financial stylized facts are explained as the finite size effects, enables the autoregulation of the number of active agents with thresholds of realized volatility for entering and exiting the markets. On the other hand, the contagion financial pricing (CFP) model [32], in which the effect of an information cascade is involved, is indeed a sophisticated version of the financial quakes model [17,33], where traders' imitating behavior of pricing is introduced by using the structure of the OFC earthquake model. Whereas basic stylized facts can be recovered with these two models, a large number of parameter values need to be specified in both models, which prevents a persuasive explanation for the universal power-law behaviors in financial markets.

### 1.2. Purpose

In this study, a simple agent-based model equipped with the SOC principle is developed for a straight exploration of the spontaneous emerging of financial stylized facts. In particular, the "Self-organized Speculation Game" (SOSG) is constructed with direct use of the structure of the BTW sandpile model. The model enables the spontaneous adjustment of market size without losing the baseline Speculation Game model's high reproducibility of stylized facts. Furthermore, the behavioral commonalities between the BTW sandpile model and the SOSG show the possibility that SOqC may work behind the spontaneity of emergence for financial stylized facts.

The remainder of this paper is organized as follows. Section 2 describes the model, and Section 3 presents the results and discussion. Finally, Section 4 concludes.

[1] A mechanism called sweeping of instability has also been proposed by Sornette et al. [22] in which no parameter tuning is needed for the spontaneous emergence of power laws. However, it does not include the adaptive adjustment of a parameter value.

**Table 1**
Example of a strategy for $M = 3$.

| History $H(t)$ | | | Recommended action $a_i(t)$ |
|---|---|---|---|
| −2 | −2 | −2 | 1 |
| −2 | −2 | −1 | 0 |
| −2 | −2 | 0 | 0 |
| −2 | −2 | 1 | −1 |
| −2 | −2 | 2 | 1 |
| −2 | −1 | −2 | 0 |
| | ⋮ | | ⋮ |
| 2 | 2 | 2 | −1 |

## 2. Self-organized speculation game

SOSG is a sophisticated model of Speculation Game, which is featured through round-trip trading with variable volumes and mutual projections between realistic and cognitive worlds.[2] Incorporating a similar structure of slowly adding elements in the BTW sandpile model, it can naturally settle the market size.

The SOSG model starts with a vacant gamified market. At each time step $t$, a new player $i$ enters the market with her initial market wealth, $w_i(t) = 10B$, where $B$ is a parameter called board lot amount which describes the ease in placing orders with multiple quantities (see Eq. (1)). Thus, the analogy of SOSG to the BTW sandpile model can be explained as follows: the gamified market corresponds to the lattice, and the new player entering the market corresponds to the falling grain of sand.

All the $N(t)$ players in SOSG will choose an action $a_i(t)$ from three alternatives: buy ($= 1$), sell ($= -1$), or hold (idle) ($= 0$). When the player submits a buy or sell (short selling is allowed) order, the quantity of order $q_i(t)$ is decided with the flooring operator $\lfloor \cdots \rfloor$ as

$$q_i(t) = \lfloor \frac{w_i(t)}{B} \rfloor. \tag{1}$$

The opening and closing quantities are kept equal in the round-trip executed by every player. Also, when the player holds a long (short) position, further buy (sell) actions are prohibited.

The market price change $\Delta p$ is formed by applying the order imbalance equation as shown in the literature [6]

$$\Delta p = p(t) - p(t-1) = \frac{1}{N(t)} \sum_{j=1}^{N(t)} a_j(t) q_j(t). \tag{2}$$

Recall that Eq. (2) implies a sufficiently liquid market realized by the existence of capable market-makers. The initial value of market price $p(t)$ can be set arbitrary, $p(0) = 100$ ($\gg 1$) is chosen in this study for convenience. Price returns are calculated as: $r(t) = \ln p(t) - \ln p(t-1)$.

Each player refers to the quinary time series of history $H(t)$, which is composed of the quantized price movements $h(t)$ that represent recent price changes in a cartoonish way. Through a magnitude comparison between $\Delta p$ and the cognitive threshold $C$, the latter of which is a threshold value for players to recognize a big price move, the quantized price movements are determined as follows:

$$h(t) = \begin{cases} 2 \text{ (largely up)} & \text{if } \Delta p > C, \\ 1 \text{ (up)} & \text{if } C \geq \Delta p > 0, \\ 0 \text{ (stay)} & \text{if } \Delta p = 0, \\ -1 \text{ (down)} & \text{if } -C \leq \Delta p < 0, \\ -2 \text{ (largely down)} & \text{if } \Delta p < -C. \end{cases} \tag{3}$$

In this way, the history $H(t)$ contains not only information on the direction, but also information on the coarse magnitude of past price movements, which accordingly requires the evaluation of trade performance in the cognitive world (see Eqs. (4) and (5)).

As each player has a finite memory, she can only read the last $M$ digits of $H(t)$. Based on this $M$ size historical pattern, the player will look up for a recommended action $a_i(t)$ in her unique strategy table, see Table 1 as an example. Note that if the recommendation has no difference with the action taken at the opening of order, the player will hold the opened position until the accomplishment of a round-trip trade. Such a setting will realize the variable holding and idling periods for each player in SOSG.

[2] See Fig. 1 of [12] for a further understanding of the framework.

In the cognitive world of each player, there exists a cognitive price $P(t)$ besides the $p(t)$ in the actual world, which is defined as follows:

$$P(t) = P(t-1) + h(t). \tag{4}$$

The performance of a round-trip trade is evaluated by the player based on the difference in $P(t)$s at the opening and closing of positions (see Eq. (5)).

When a round-trip trade is closed with the reverse order, the cognitive gain $\Pi_i(t)$ is firstly computed in players' cognitive world, mimicking how humans handle numerical information in a heuristic manner (e.g., the precision effect and the ease of computation effect mentioned in Thomas and Morwitz's study [34]). When player $i$ opens a position at $t_0$ and closes it at $t$, her cognitive gain per unit quantity of order can be found as

$$\Pi_i(t) = a_i(t_0)(P(t) - P(t_0)) = a_i(t_0) \sum_{k=t_0+1}^{t} h(k). \tag{5}$$

After the evaluation, the player will update her market wealth in the realistic world, through the investment adjustment $\Delta w_i(t)$ determined by the conversion of $\Pi_i(t)$ with the consideration of her trading volume $q_i(t)$:

$$w_i(t) = w_i(t_0) + \Delta w_i(t) = w_i(t_0) + f(\Pi_i(t) q_i(t_0)). \tag{6}$$

where $f$ can be an arbitrary monotonically increasing function. For simplicity, $\Delta w_i(t) = \Pi_i(t) q_i(t_0)$ is used in the SOSG. Note that this $\Delta w_i(t)$ would be different from the gain or loss calculated directly based on the market price, though the difference is assumed to be adjusted externally; in other words, self-finance is not postulated in the SOSG. Lastly, if the wealth of a player decreases to the level $w_i(t) < B$, making her impossible to order even one unit of quantity, the player will be forced to leave the market. Hence, $B$ also works as a significant threshold value to induce irregular withdrawals of players, which is analogous with the fact that the threshold dynamics play an important role in the BTW sandpile model.

To emphasize, the major difference in the structure of SOSG from that of the standard Speculation Game is the reduction of two main control parameters: the number of players $N$ and the number of players' strategies $S$. Instead of the fixed population size with substitutions for bankruptcies in the Speculation Game, $N$ in the SOSG can change and eventually be settled through a constant inflow of a player to a gamified market and irregular outflows of bankrupted players from it. Additionally, the value of $S$ is set as one, considering the finding in our previous study [13] that inductive learning is not indispensable to the reproduction of stylized facts with strategy tables in the Speculation Game.

## 3. Results and discussions

### 3.1. Market size self-adjustment

First, we summarize that the market size can be spontaneously adjusted in SOSG along with the reproduction of volatility clustering in the price returns. Figs. 1 and 2 display the time series of the number of players participating in the gamified market and the corresponding price returns $r(t)$, respectively. The three main parameters of the SOSG are set as $M = 5$, $B = 9$, and $C = 3$ by following the baseline setting of the Speculation Game in [12] and [13]. Henceforth, the memory size and the cognitive threshold are fixed as $M = 5$ and $C = 3$. In Fig. 1, the number of players surges at the beginning of the simulation and subsequently fluctuates around approximately 700 players. In the meantime, Fig. 2 shows that the magnitudes of $r(t)$ fluctuate intermittently, indicating the stable emergence of volatility clustering.

Note that the SOSG is equivalent to the original Speculation Game [12] regarding the reproducibility of stylized facts (10 out of 11 stylized facts in Cont's report [1] are confirmed except for the gain/loss asymmetry [35]). Additionally, the emerging mechanism of volatility clustering and the effects of the parameters in SOSG are identical to those elucidated in [13] as well. Furthermore, similar to the Perturbated Speculation Game [36], the introduction of perturbation to the SOSG price change, which represents the effects of non-pure speculators, can generate price trends so that the correct Hurst exponent can be obtained; thus the long-term scaling of fluctuation is ensured. Therefore, by maintaining the basic features of the Speculation Game, the relaxation in the requirement of a predetermined market size suggests that SOSG can be considered a smarter model that abstracts more essential elements of the financial market.

### 3.2. Commonalities with the BTW sandpile model

#### 3.2.1. Spontaneous settlement of system size

In the following sections, we show that the behavior of the BTW sandpile model and SOSG are quite similar. First, both models have the characteristic that the final system size, to which the required time length for adjustment is proportional, is highly determined by a single control parameter. In the BTW sandpile model, the number of sand grains that remain on the lattice depends on the length of its side $L$ as indicated in Fig. 3. Larger $L$ does not only increase the number of retained sand grains but also lengthens necessary time steps to reach stable states. Similarly in the SOSG, the number of players staying in the gamified market strongly hinges on the value of the board lot amount $B$. A larger $B$, implying more difficult market conditions for players to place large orders, results in a larger market size and a longer time for convergence, as indicated in Fig. 4. However, the fluctuations in leaving players in SOSG during the stable state are a bit larger than those in the outflowing sand grains in a BTW sandpile model.

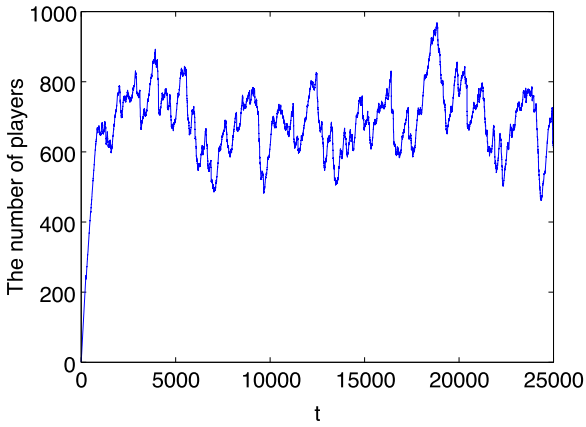


**Fig. 1.** Dynamic number of players in the gamified market.

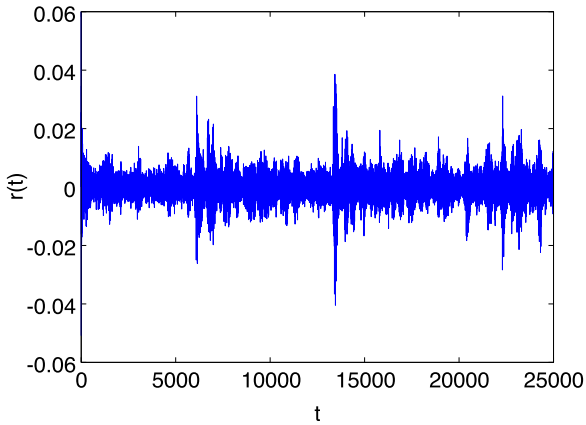


**Fig. 2.** Intermittent time series of price returns corresponding to Fig. 1.

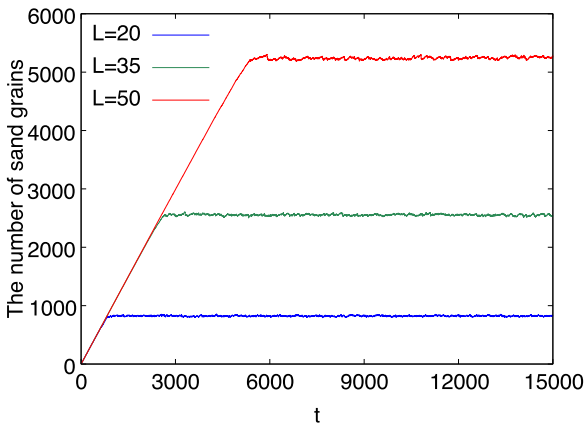


**Fig. 3.** Time series of number of sand grains remaining on the lattice with different $L$ in BTW sandpile model.

### *3.2.2. Power laws and quasi-criticality*

Moreover, similar to the relationship between the frequency and size of sand avalanches in the BTW sandpile model, power-law distributions of several statistical quantities can be observed in SOSG.

One such distribution is that of the heavy tails of price returns, one of the fundamental stylized facts previously explained. The blue line in Fig. 5 shows a good agreement to the power law for the complementary cumulative distribution of positive market returns generated from 20-trial in SOSG simulations of 50,000-time steps. Its power-law feature is

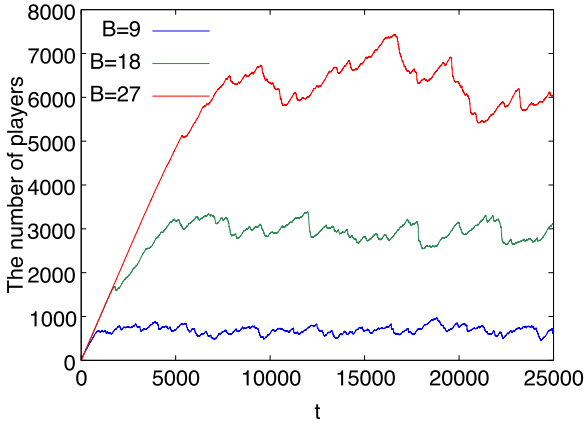


**Fig. 4.** Time series of number of players participating in the gamified market with different $B$ in SOSG.

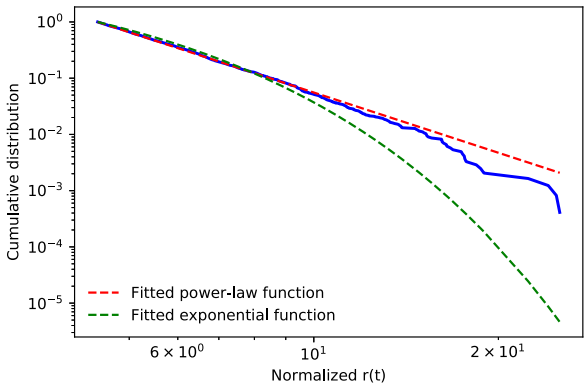


**Fig. 5.** Complementary cumulative distribution of positive market returns (500,195 data points generated by 20 simulation runs) when $B = 9$.

examined using the Python package [37], with which one can implement the robust statistical analysis advocated by Clauset, Shalizi, and Newman [38]. The red dashed line of the estimated power-law function with a tail index of $\alpha = 4.56$, satisfying the range condition of empirical exponents: $2 < \alpha < 5$, visually fits the blue data line better than the green dashed line of the estimated exponential function. Further, Vuong's test [39] also concluded that the power-law distribution is closer to the genuine one because the likelihood ratio $\mathcal{LR}$ is $3.54(> 0)$ and the $p$-value is $4.05 \times 10^{-4}(< 0.1)$. Note that the cumulative distribution of negative market returns and those in the cases of different $B$ values also exhibit the same power-law tails.

Another power-law statistic is the cumulative distribution of $\lfloor \Delta w/B \rfloor$, the investment adjustment per unit board lot amount that relates to a player's earns through a round-trip trade or, more simply, the capital change in terms of the order quantity (see Eq. (1)). This statistical quantity can be explained as the amount of market wealth exchanged among the players, which intuitively corresponds to the number of sand grains involved in an avalanche cascading in the BTW sandpile model. Nonetheless, as the direct relationship of "a player = a sand grain" exists; such an intuitive analogy is not fully supported. The circular points drawn in Fig. 6 with the use of R version of the power-law package [40] show the complementary cumulative distribution of positive $\lfloor \Delta w/B \rfloor$ obtained from the round-trip trades done by all players in a single simulation of 50,000-time steps. The statistical analyses of Clauset et al. [38] gives a better fitted power-law function (shown as the red solid line in the figure) with a tail index $\alpha = 3.38$ than the exponential one (shown as the green dashed line in the figure). Because the likelihood ratio and the $p$-value of Vuong's test [39] are $\mathcal{LR} = 7.18(> 0)$ and $p = 7.05 \times 10^{-13}(< 0.1)$, respectively, a closer fitting of the power-law function is quantitatively ensured as well. Such a power-law tail is also held for the cumulative distribution of negative $\lfloor \Delta w/B \rfloor$ and for different $B$, similar to the distributions of price returns explained in the previous section.

Whereas SOSG can also reproduce a long time tail in the autocorrelation of volatilities, the decay of autocorrelation does not follow the power-law function. Indeed, the decay can be better regressed to a logarithmic function, which is similar to the simulation results of the standard Speculation Game [12,13]. Fig. 7 displays three cases of the autocorrelation $\rho_v(\tau)$ of volatilities $|r(t)|$, where $\rho_v(\tau)$ is calculated using the market price returns for 50,000-time steps and averaged over 100 trials, for $B = 9$, 18, and 27. The absence of the power-law property indicates that the infinite correlation over time is not

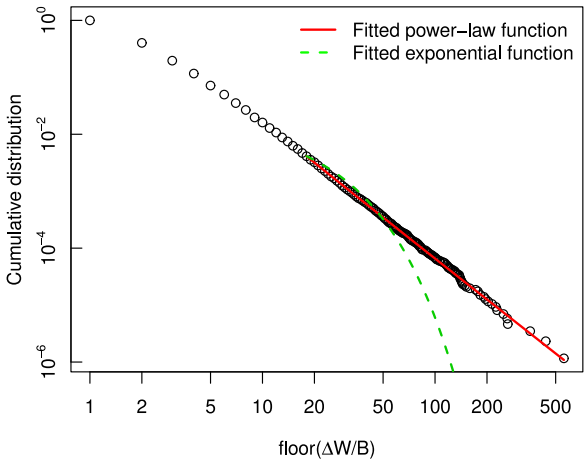


**Fig. 6.** Complementary cumulative distribution of $\lfloor \Delta w/B \rfloor (> 0)$ (860,995 data points from a single simulation) when $B = 9$.

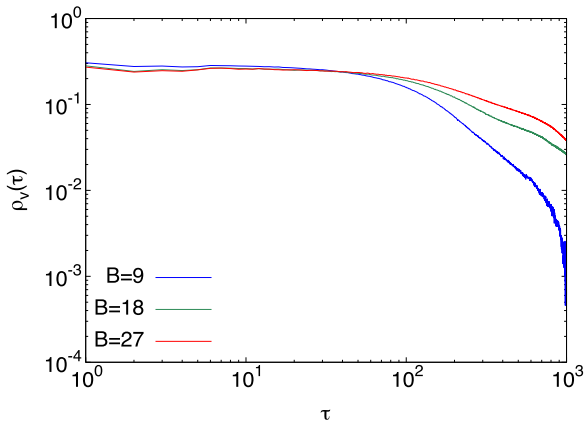


**Fig. 7.** Slow decays of autocorrelation in volatility with various $B$ in SOSG.

established in SOSG. Therefore, we reached the conclusion that eventually the SOSG system settles in a near quasi-critical rather than a purely critical state. This justifies what we mentioned in the introduction section that, the SOSG is a toy model for the financial markets, equipped with the SOqC mechanism.

### *3.3. Reasons for spontaneous market-size settlement*

The spontaneous settlement of system size under SOC in the BTW sandpile model has been explained with a general concept called the absorbing phase transition, which is presented in the theory of phase transition for non-equilibrium systems [41]. The absorbing phase transition is defined as a systematic changeover from the active phase, during which the reaction never stops and continuously changes the structure, to the absorbing state, namely, a state in which the system is trapped forever with a monotonous structure once it is reached. Because the separation of timescales in the BTW sandpile model allows the unlimited dissipation of sand grains, the particle density on the lattice is ensured to recover its critical value for which the absorbing phase transition occurs even though the slow addition of grain will drive the system back to the active phase. Such a mechanism leading the system to the critical point acts as an attractor for the system dynamics, which can stabilize the number of sand grains remaining on the lattice.

Intuitively, the size settlement in SOSG should also have some relationship with the absorbing phase transition. However, the direct explanation shown above cannot apply to SOSG, because the "lattice" size of the gamified market that serves as a physical space for the residence of SOSG players is not fixed. In contrast, the lattice that similarly functions as a receiver of sand grains in the BTW sandpile model is invariant. Furthermore, if the exchange or redistribution of wealth among the SOSG player could be loosely defined as a kind of local "dissipation", surely the rate of such a dissipation would not be unlimited due to the lack of catenated responses. Hence, we can observe larger fluctuations in the time evolution of that number of players than that of sand grains.

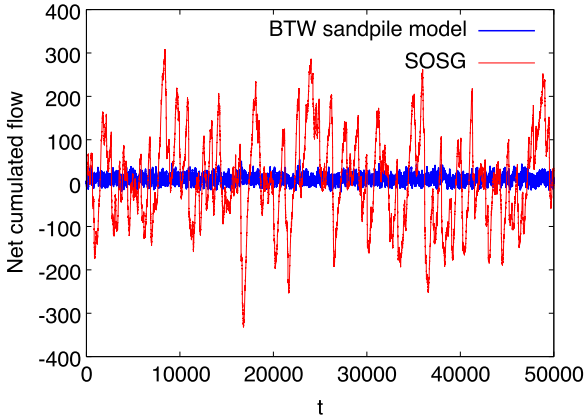


**Fig. 8.** Net cumulative numbers of inflowed and outflowed sand grains on the lattice of BTW sandpile model ($L = 20$) and entering and exiting players in the gamified market of SOSG ($B = 9$) in the stable state. The transition periods are considered as 1,000-time steps and 2,000-time steps, respectively.

Here, we explicate the reason, from a different aspect, for why SOSG can keep the market size to a certain fixed level and drive the system near to criticality in an averaged sense. In particular, we show the balance of the time-averaged inflow and outflow by taking the net statistics of both cumulative elements in the stable state. In the case of the BTW sandpile model, the difference between the total amounts of sand grains flowing into and out of the lattice intensively fluctuates around zero, shown as the blue line in Fig. 8. This implies the cumulation of the runoff through the avalanches grows almost identically as to the total deposit of sand grain, which is the accumulation of the falling sand particle at each time step. Therefore, the two flows averaged in time almost match each other as

$$\langle N_{net} \rangle = \frac{1}{t}\left(\sum_t N_{in} - \sum_t N_{out}\right) \approx 0 \quad \Leftrightarrow \quad \langle N_{in} \rangle \approx \langle N_{out} \rangle \ (\approx 1), \tag{7}$$

where $N_{net}$, $N_{in}$, and $N_{out}$ are the numbers of net flows, inflows, and outflows of sand grains at each time step $t$, respectively. The balance of the averaged inflow and outflow indicated by such simple statistics assures the convergence of the amount of remaining sand on the lattice in the BTW sandpile model.

A similar balancing phenomenon occurs in the SOSG system. The red line in Fig. 8 shows the time series of the difference between the total number of entering and exiting players in the gamified market. Although it vibrates larger than the blue line does, which reflects the nontrivial fluctuation in the number of outflowing players as Figs. 1 and 4 suggest, its mean-reverting feature to zero can be observed. This connotates that the accumulation of withdrawn players inflates almost equivalently to the total population of players entering at each time step. Accordingly, similar to the case of the BTW sandpile model, Eq. (7) can be established in the SOSG system as well, designating that the time-averaged number of players entering and exiting are balanced with each other.

Moreover, the dynamic equilibrium regarding the number of players remained in the gamified market is a superficial feature, the essence indeed lies in the flow of market wealth. Three curves depicted in Fig. 9 demonstrate the cumulative changes in market wealth caused by the inflow of wealth through a single route and the outflow of wealth through two different routes. In the SOSG model setting, the entering players bring $10B$ wealth into the gamified market at each time step; naturally, its total amount increases monotonically as a linear function of time (shown as the blue line in Fig. 9). The inflowed market wealth is redistributed among the players through round-trip trades. In the redistribution process, most of the market wealth will be deprived by the market makers, whose existence is required by applying the excess demand to the calculation of the price change (in Eq. (2)). The acquired wealth of the market maker (shown as the red line in Fig. 9) can be calculated by executing holding positions in chronological order. Because the SOSG players utilize an extended strategy table of the Minority Game type, the players have little chance to continuously win trades in the long term, a feature that is similar to the previous Minority Game [7] and the $-game [42]. Eventually, the market maker will win alone, albeit with some nontrivial loss, indicated by occasional dips. As the second route of outflow, some market wealth is also brought out of the gamified market by the bankrupted players. The cumulative amount of this outflow grows slowly and linearly as well (shown as the green line). Finally, the sum of these two cumulative outflows equals the accumulated market wealth that flowed into the market.

Hence, in the time averaging sense, a dynamic equilibrium state can be established in the SOSG system through the balance between the total inflow and outflow of market wealth as

$$\begin{aligned} &\sum_t w_{in} \approx \sum_t w_{maker} + \sum_t w_{exit} \approx \sum_t w_{out} \\ \Leftrightarrow \quad &\langle w_{in} \rangle \approx \langle w_{maker} \rangle + \langle w_{exit} \rangle \approx \langle w_{out} \rangle \ (\approx 10B), \end{aligned} \tag{8}$$

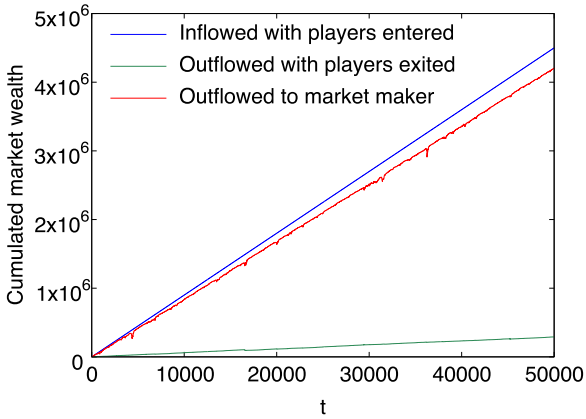


**Fig. 9.** Cumulative amounts of inflowed and outflowed market wealth $w$ in the gamified market of SOSG. The wealth that flowed into the market (the blue line) was brought by entering players partially flows away from the market by the leaving players (the green line), and most of it flows into the market maker (the red line).

where $w_{in}$, $w_{maker}$, $w_{exit}$, and $w_{out}$ are the wealth amounts of the inflow, the outflow to the market maker, the outflow with exiting players, and the total outflow at each time step, respectively. Moreover, the relationship between the red and the green lines in Fig. 9 implies

$$\frac{\langle w_{exit} \rangle}{\langle w_{out} \rangle} = \frac{1}{O(1)}, \tag{9}$$

which means that, in the time average sense, if each time a bankrupted player left the market with market wealth $\langle w_{exit} \rangle$, the total outflow $\langle w_{out} \rangle = O(1)\langle w_{exit} \rangle$ would balance with the inflow of $\langle w_{in} \rangle$ brought by a player newly entering the market. Consequently, an equation similar to Eq. (7) can also be established, showing the balance in the averaged number of entering and exiting players.

## 4. Conclusion

In this paper, the SOC mechanism is incorporated into the Speculation Game for the realization of spontaneous adjustments of the market size in the process of generating stylized financial properties. Compared with the original model, the established SOSG succeeds in reducing the number of main parameters from five ($N$, $S$, $M$, $B$, $C$) to three ($M$, $B$, $C$) and maintains the high reproducibility of 10 stylized facts. In this sense, the SOSG model can be regarded as an essential type of Speculation Game. Moreover, our study reveals that SOSG shares several common phenomenological characteristics with the BTW sandpile model: the spontaneous settlement of system size, distributions following the power laws, and the balance (in the time-averaging sense) between system inflows and outflows. On the other hand, the SOSG system can reach merely a subcritical state in reference to the non-power-law decay of the autocorrelation in volatilities. Additionally, the stabilizing mechanism of system size in the SOSG is slightly more complex because the averaged number of inflowing and outflowing players in the gamified market is essentially related to the averaged amount of inflowing and outflowing market wealth behind.

Results of our study suggest the potential existence of SOqC in the actual financial markets as well as its possible contribution to the spontaneous emergence of stylized facts. The two essential premises here are that the financial markets are open systems and that capital flows through the entrance and the exit of market participants. It is also notable that SOSG does not require the spacial structure, or local interaction, for the emergence of SOqC, which is significantly different from the settings of the BTW sandpile model.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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